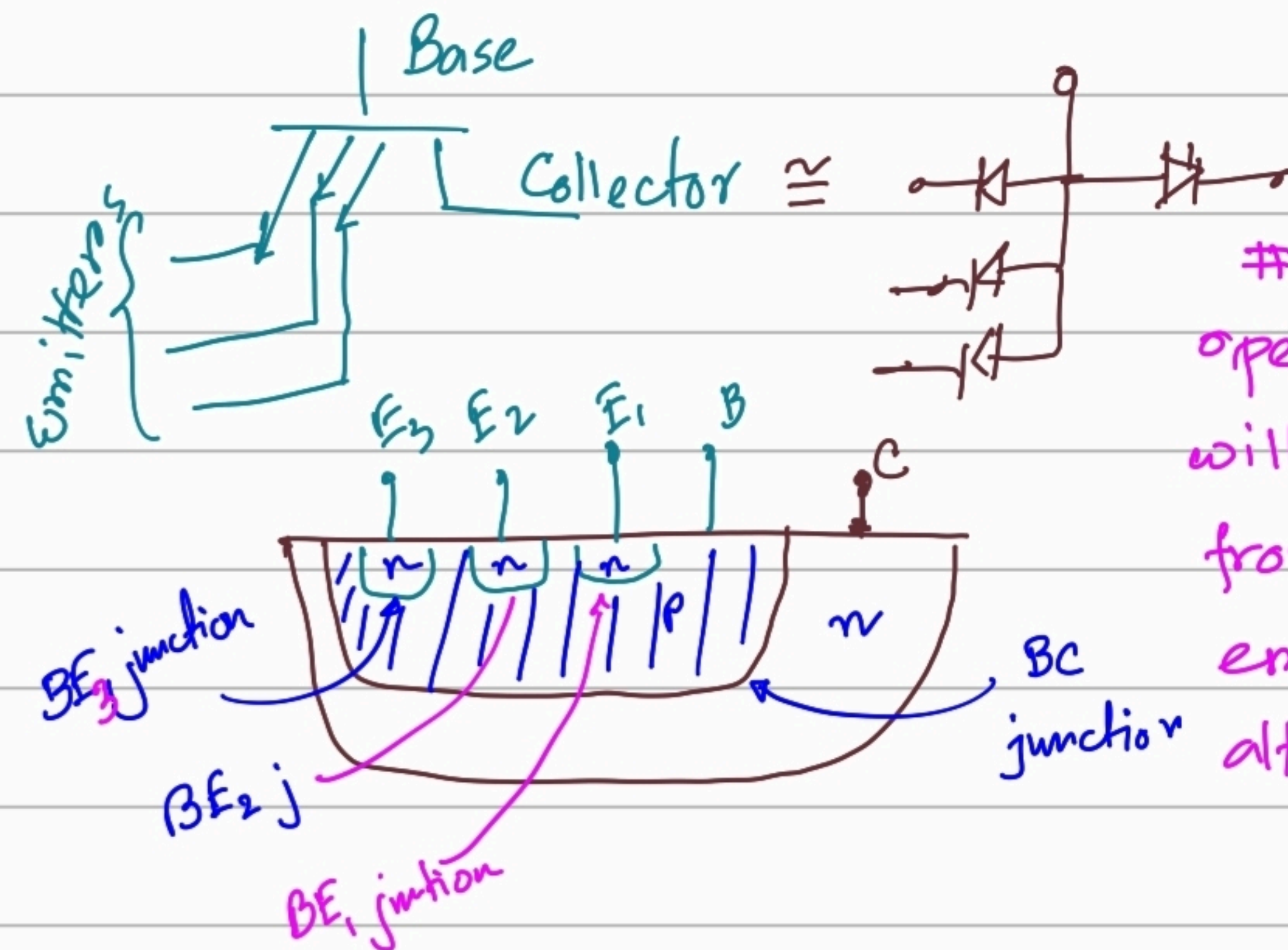


"Transistor-Transistor Logic"

Bipolar, $\rightarrow$ Saturated $\rightarrow$ TTL

BJT Switching time is faster than DTL.

- # The input terminals are connected to the emitter terminal of a transistor and the output terminal is connected to a switching transistor.
- # input terminals will be connected to multi-emitter transistor.



Transistor's operating mode will be determined from three emitter inputs altogether.

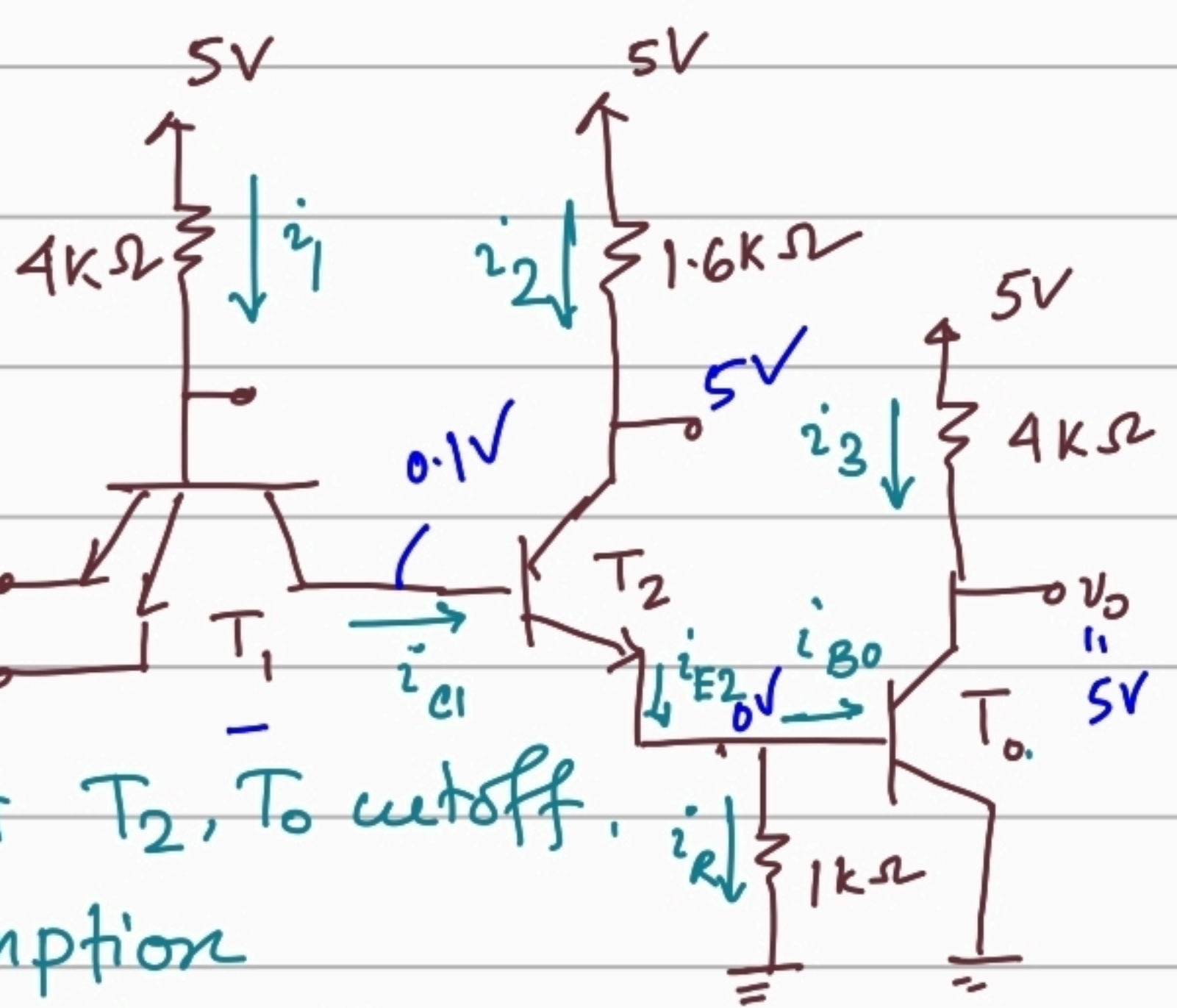
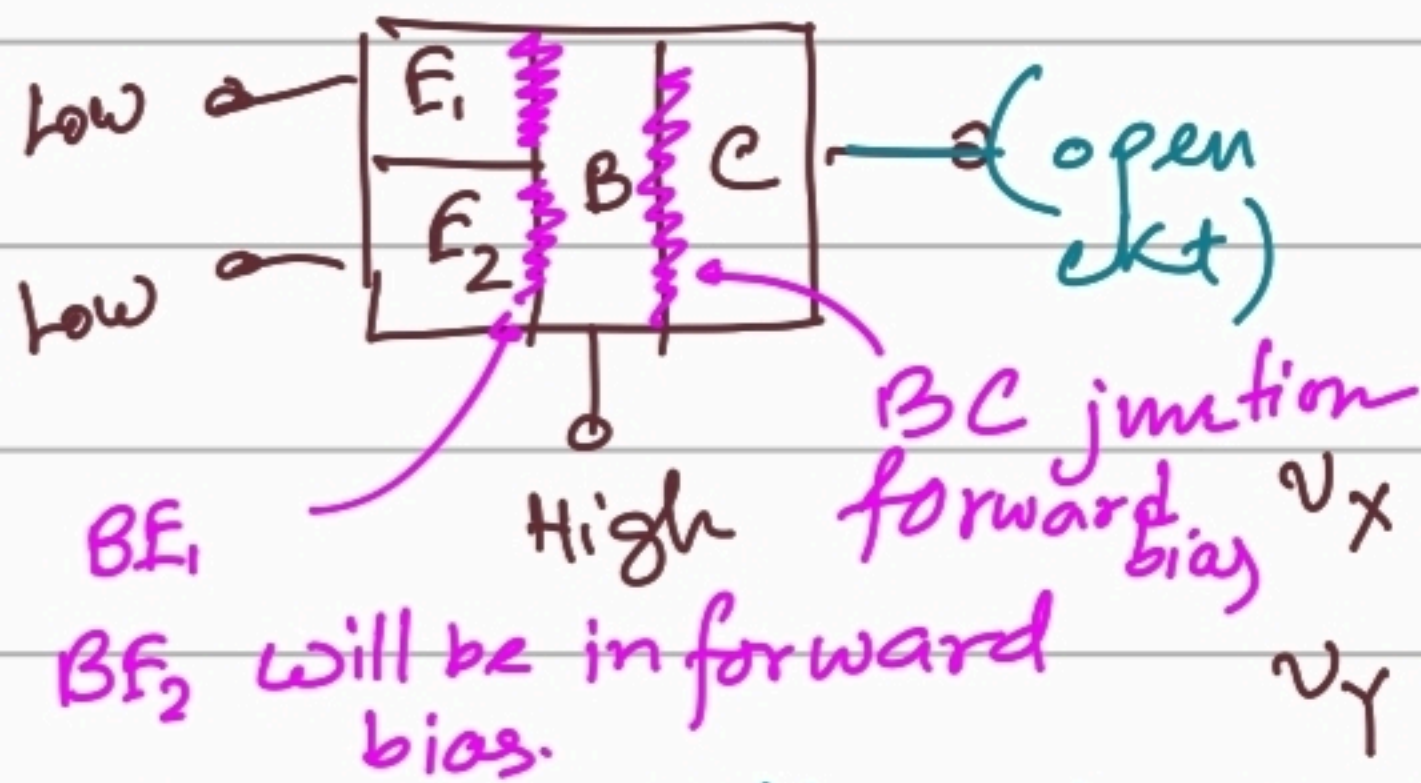
Basic Operation:

Find out the current and voltage of the following gate. Verify that it works like a NAND gate.

Case ①. $\underline{v_x = v_y = 0.1V}$

[Neaman] $v_{CE}(\text{sat}) = 0.1V$
 $v_{BE}(\text{sat}) = 0.8V$

For T_1 transistor



Our assumption is that T_2, T_3 cutoff.

According to our assumption

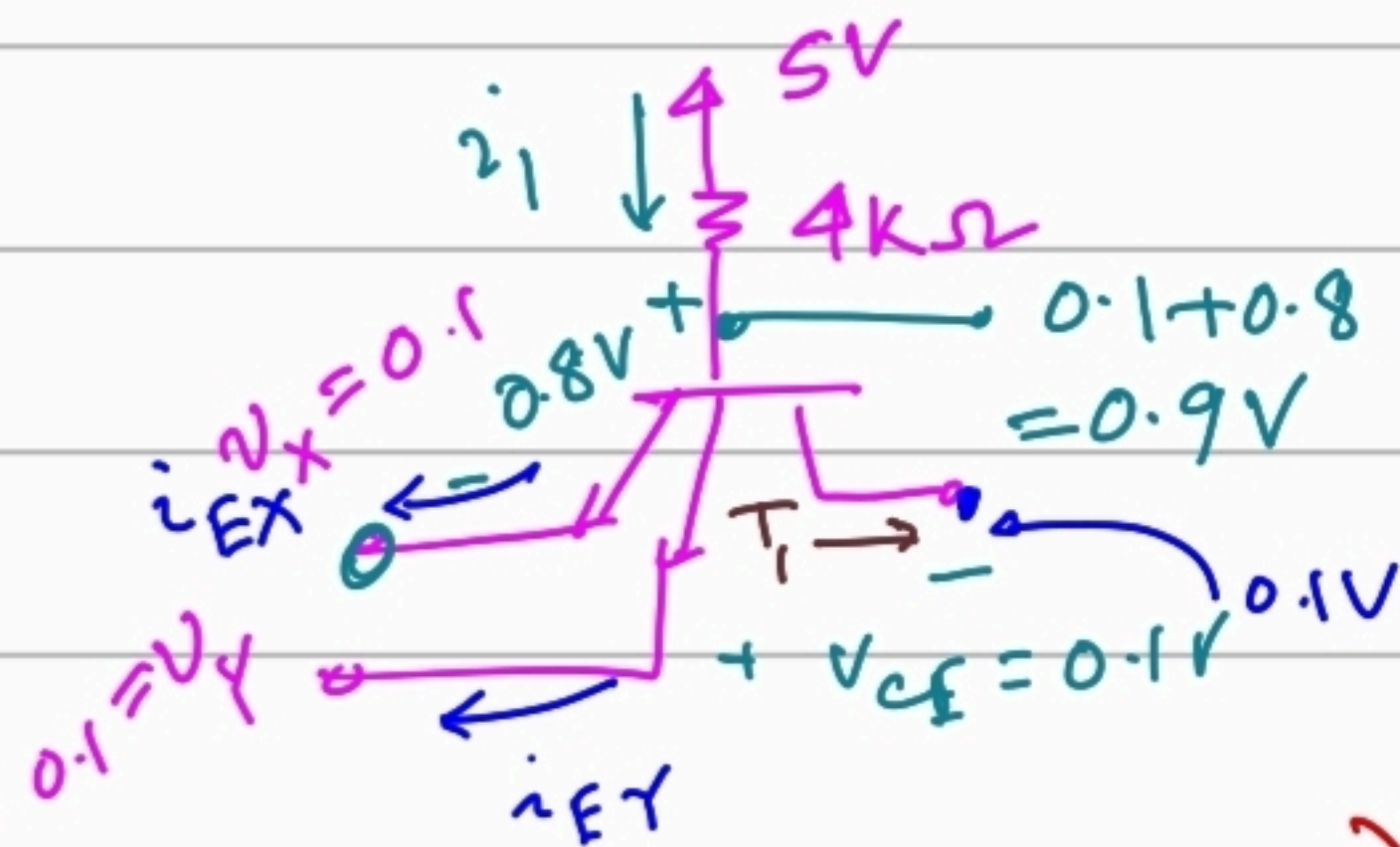
$$i_2 = i_3 = i_R = i_{B0} = i_{E2} = i_{C1} = 0 \text{ mA}$$

Bonus Question: B.C terminal's bias?

Ans: If B.C reverse bias, T_1 will be in forward active but for forward active we need, $i_C = \beta i_B$. But $i_C = 0$. Therefore not possible.

Hence T_1 must be in saturation mode. $\Rightarrow$

BC-junction is in forward bias

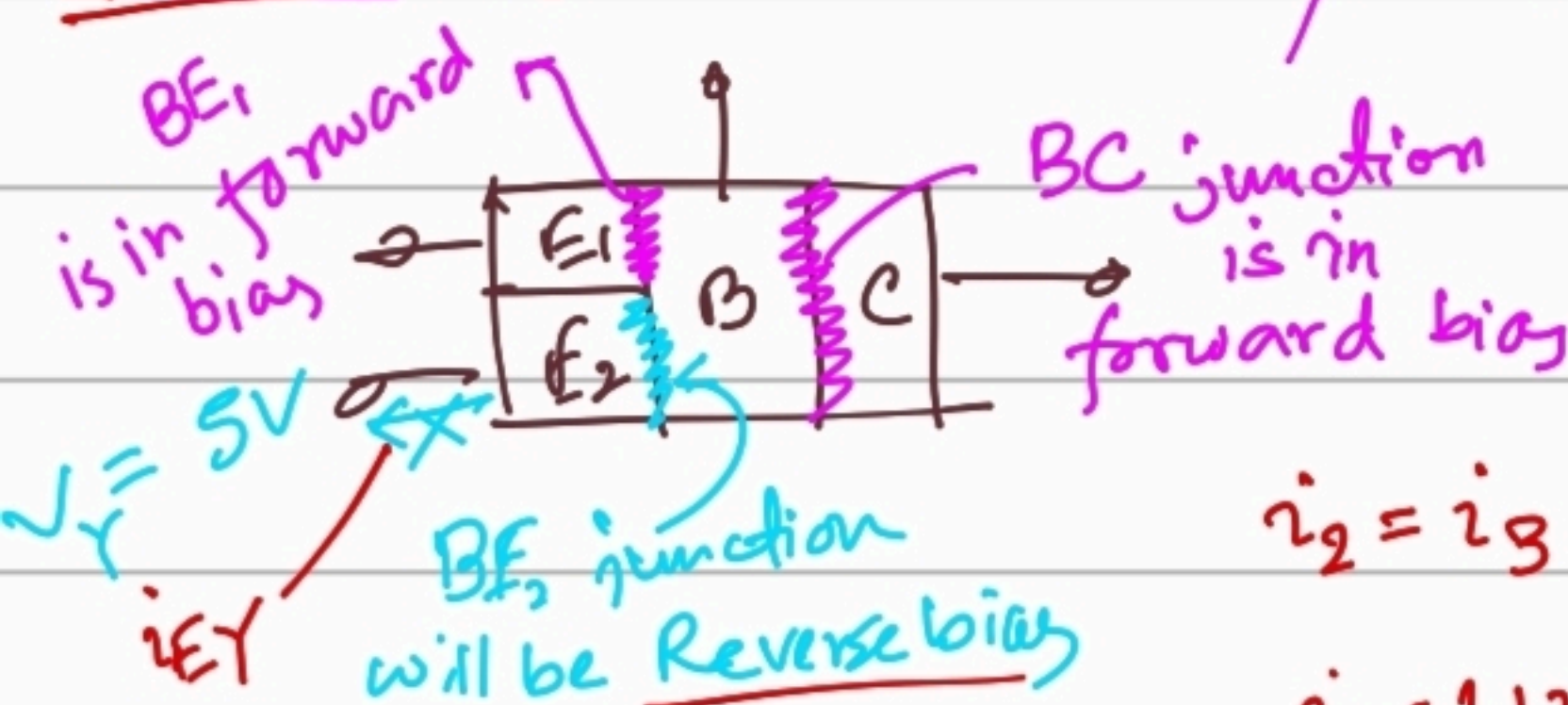


$$i_1 = \frac{5 - 0.9}{4 \text{ k}\Omega} = 1.125 \text{ mA}$$

$$i_{EX} = i_{EY} = \frac{i_1}{2} = 0.5675 \text{ mA}$$

$$\beta_{\text{forced}} = \frac{i_C}{i_B} = 0 < \beta_F \text{ Our assumption is valid.}$$

Case ② ($V_X = 0.2 \text{ V}, V_Y = 5 \text{ V}$) / $V_Y = 0.2 \text{ V}, V_X = 5 \text{ V}$



$T_1 \rightarrow$ saturation.

$T_2, T_3 \rightarrow$ cutoff.

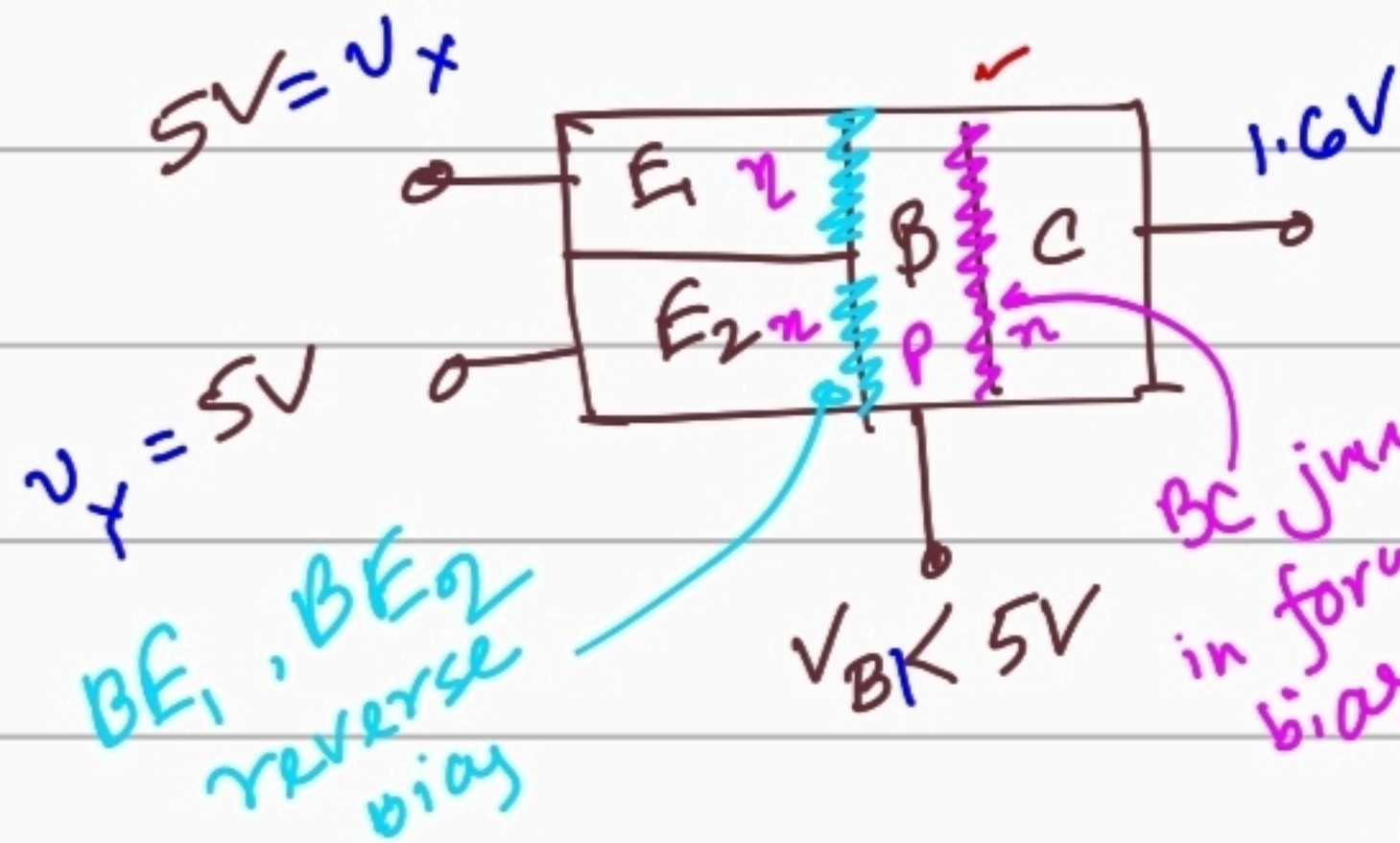
$$i_2 = i_3 = i_{B0} = i_{E2} = i_{C1} = i_R = 0$$

$$i_1 = 1.125 \text{ mA. } i_{EX} = i_1, i_{EY} = 0$$

Case 3: $v_x = v_y = 5V$ | Assumption:

$T_0, T_2 \rightarrow$ saturation

$$v_{CE}(\text{sat}) = 0.1V, v_{BE}(\text{sat}) = 0.8V$$



BC junction is in forward bias

$$\beta_F = 25$$

$$\beta_R = 0.1$$

T_1 transistor must be in reverse active mode.

Collector and Emitter terminals will switch their roles.

$V_{BC}(\text{reverse active}) = 0.7V$. [because BC junction is in forward bias]

$(i_{EX} = \beta_R i_B = i_{EY})$ β_R = reverse active current gain.

$$i_{B1} = i_1 = \frac{5 - 2.3}{4k} = 0.675 \text{ mA}, i_2 = \frac{5 - 0.9}{1.6k} = 2.5625 \text{ mA}$$

$$i_3 = \frac{5 - 0.1}{4k} = 1.225 \text{ mA}, i_{EX} = i_{EY} = \beta_R i_{B1} = 0.0675 \text{ mA}$$

$$i_{C1} = i_{EX} + i_{EY} + i_{B1} = 0.81 \text{ mA}, i_{E2} = i_2 + i_{C1} = 3.3725 \text{ mA}$$

$$i_R = \frac{0.8 - 0}{1k} = 0.8 \text{ mA}, i_{B0} = i_{E2} - i_R = 2.5725 \text{ mA}$$

$$\beta_{\text{forced}}(T_2) = \frac{i_2}{i_{C1}} = \frac{2.5625}{0.81} = 3.16 < \beta_F = 25$$

$$\beta_{\text{forced}}(T_0) = \frac{i_3}{i_{B0}} = \frac{1.225}{2.5725} \approx 0.47 < \beta_F = 25$$

Our assumptions are valid.

v_x	v_y	v_o
0.1	0.1	5
0.1	5	5
5	0.1	5
5	5	0.1

This circuit works like a NAND gate.